

# CS & IT ENGINEERING



## Data Structure & Programming

### Stack and Queues

Lecture No.- 06



By- Satya sir

# Recap of Previous Lecture



Simple Queue

Circular Queue

Double Ended Queue

Priority Queue

Enqueue() and

Dequeue() Operations

- Stack Implementation Using Linked Lists
  - Push  $\cong$  Insertion at Beginning
  - Pop  $\cong$  Deletion of first node
- Queue Implementation Using Linked Lists
  - Enqueue()  $\cong$ 
    - Insertion at front (or)
    - Insertion at last
  - Dequeue()  $\cong$ 
    - Deletion of last (or)
    - Deletion of first

# Topics to be Covered



- Stack Implementation
  - Using Queues
- Queue Implementation
  - Using Stacks





## Topic : Stack Implementation using Queues



### Stack Implementation Using Queues

- 2 Queues are required.

Let  $Q_1$  and  $Q_2$  Initially Empty

- When we Push Elements A, B, C, D, E in the same order,  
Expected Pop (or) Sequence : E, D, C, B, A if Used Stack.

- If Used Queue, with Insertion in the order A, B, C, D, E  
The o/p Sequence would be : A, B, C, D, E

Stack Can be implemented Using 2 Queues  
in Either of 2 Ways :

- ① By making Push operation Costly

-  $\text{Push}() \Rightarrow O(n)$

-  $\text{Pop}() \Rightarrow O(1)$

- ② By making Pop operation Costly

-  $\text{Push}() \Rightarrow O(1)$

-  $\text{Pop}() \Rightarrow O(n)$



# Topic : Stack Implementation using Queues

Let Operations need to be Performed are :

Push(A), Push(B), Push(C), Pop(), Pop(),  
Push(D), Pop(), Push(E), Pop(), Pop()

Expected o/p: C, B, D, E, A



Push operation Costly

Push() :  $O(n)$

Step ① Dequeue() all Elements from  $Q_1$  and Enqueue() into  $Q_2$ . ( $Q_2 \leftarrow Q_1$ )

Step ② Enqueue() Current Element into  $Q_1$ . ( $Q_1 \leftarrow \text{value}$ )

Step ③ Dequeue all Elements from  $Q_2$  and Enqueue back into  $Q_1$ . ( $Q_1 \leftarrow Q_2$ )

Pop() :  $O(1)$

Step : Dequeue() from  $Q_1$  and return as o/p.

|                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><u>Push(A)</u> <math>Q_1 \square Q_2 \square</math></p> <ol style="list-style-type: none"> <li><math>Q_2 \leftarrow Q_1</math> —</li> <li><math>Q_1 \leftarrow A</math> <math>Q_1 [A] Q_2 \square</math></li> <li><math>Q_1 \leftarrow Q_2</math> — —</li> </ol>                                                               | <p><u>Pop()</u> <math>Q_1 [C B A] Q_2 \square</math></p> <p>Dequeue from <math>Q_1 \Rightarrow</math> <u>O/p: C</u></p> <p><math>Q_1 [B A] Q_2 \square</math></p> <hr/> <p><u>Pop()</u> : Dequeue from <math>Q_1 \Rightarrow</math> <u>O/p: B</u></p> <p><math>Q_1 [A] Q_2 \square</math></p>                                                                      |
| <p><u>Push(B)</u> <math>Q_1 [A] Q_2 \square</math></p> <ol style="list-style-type: none"> <li><math>Q_2 \leftarrow Q_1</math> <math>Q_1 \square Q_2 [A]</math></li> <li><math>Q_1 \leftarrow B</math> <math>Q_1 [B] Q_2 [A]</math></li> <li><math>Q_1 \leftarrow Q_2</math> <math>Q_1 [B A] Q_2 \square</math></li> </ol>         | <p><u>Push(D)</u></p> <ol style="list-style-type: none"> <li><math>Q_1 \square Q_2 [A]</math></li> <li><math>Q_1 [D] Q_2 [A]</math></li> <li><math>Q_1 [D A] Q_2 \square</math></li> </ol> <hr/> <p><u>Pop()</u> : <u>O/p: D</u></p> <p><math>Q_1 [A] Q_2 \square</math></p> <hr/> <p><u>Pop()</u> : <u>O/p: A</u></p> <p><math>Q_1 \square Q_2 \square</math></p> |
| <p><u>Push(C)</u> <math>Q_1 [B A] Q_2 \square</math></p> <ol style="list-style-type: none"> <li><math>Q_2 \leftarrow Q_1</math> <math>Q_1 \square Q_2 [B A]</math></li> <li><math>Q_1 \leftarrow C</math> <math>Q_1 [C] Q_2 [B A]</math></li> <li><math>Q_1 \leftarrow Q_2</math> <math>Q_1 [C B A] Q_2 \square</math></li> </ol> | <p><u>Push(E)</u> :</p> <ol style="list-style-type: none"> <li><math>Q_1 \square Q_2 [A]</math></li> <li><math>Q_1 [E] Q_2 [A]</math></li> </ol> <hr/> <p>obtained o/p:<br/><u>C, B, D, E, A</u></p>                                                                                                                                                               |



# Topic : Stack Implementation using Queues

Let Operations be :

Push(A), Push(B), Push(C), Pop(), Pop(), Push(D),  
Pop(), Push(E), Pop(), Pop()



Pop() operation costly

Push() :  $O(1)$

1. Enqueue into  $Q_1$

Pop() :  $O(n)$

Step 1: Dequeue all Elements from  $Q_1$   
except last Element and Enqueue into  $Q_2$ .

Step 2: Dequeue last Element from  $Q_1$  and return as o/p.

Step 3: Dequeue all Elements from  $Q_2$  and  
Enqueue back to  $Q_1$ .

Expected o/p sequence : C, B, D, E, A

Push(A)  $Q_1$   $\square$   $Q_2$   $\square$

$Q_1 \leftarrow A$   $Q_1$   $\boxed{A}$   $Q_2$   $\square$

Push(B)  $Q_1 \leftarrow B$

$Q_1$   $\boxed{A|B}$   $Q_2$   $\square$

Push(C)  $Q_1 \leftarrow C$

$Q_1$   $\boxed{A|B|C}$   $Q_2$   $\square$

Pop()  $Q_1$   $\boxed{A|B|C}$   $Q_2$   $\square$

1.  $Q_1$   $\boxed{C}$   $Q_2$   $\boxed{A|B}$

2. O/p: C  $Q_1$   $\square$   $Q_2$   $\boxed{A|B}$

3.  $Q_1 \leftarrow Q_2$   $Q_1$   $\boxed{A|B}$   $Q_2$   $\square$

Pop()  $Q_1$   $\boxed{A|B}$   $Q_2$   $\square$

1.  $Q_1$   $\boxed{B}$   $Q_2$   $\boxed{A}$

2. O/p: B  $Q_1$   $\square$   $Q_2$   $\boxed{A}$

3.  $Q_1 \leftarrow Q_2$   $Q_1$   $\boxed{A}$   $Q_2$   $\square$

Push(D) :  $Q_1 \leftarrow D$   $Q_1$   $\boxed{A|D}$   $Q_2$   $\square$

Pop() 1.  $Q_1$   $\boxed{D}$   $Q_2$   $\boxed{A}$

2. O/p: D  $Q_1$   $\square$   $Q_2$   $\boxed{A}$

3.  $Q_1$   $\boxed{A}$   $Q_2$   $\square$

Push(E) :  $Q_1 \leftarrow E$  :  $Q_1$   $\boxed{A|E}$   $Q_2$   $\square$

Pop() 1.  $Q_1$   $\boxed{E}$   $Q_2$   $\boxed{A}$

2. O/p: E  $Q_1$   $\square$   $Q_2$   $\boxed{A}$

3.  $Q_1$   $\boxed{A}$   $Q_2$   $\square$

Pop() 1.  $Q_1$   $\boxed{A}$   $Q_2$   $\square$

2. O/p: A  $Q_1$   $\square$   $Q_2$   $\square$

3. —

Obtained o/p: C, B, D, E, A



## Topic : Queue Implementation using Stack



- To implement Queue Using Stack, 2 stacks are Used.

- 2 Ways :

① Enqueue() operation costly

- Enqueue() :  $O(n)$

- Dequeue() :  $O(1)$

② Dequeue() operation costly

- Enqueue() :  $O(1)$

- Dequeue() :  $O(n)$



# Topic : Queue Implementation using Stack



Enqueue() operation costly : Let 2 stacks  $S_1, S_2$  initially Empty.

Enqueue() :  $O(n)$

- ① Pop all Elements from  $S_1$  and Push to  $S_2$ . ( $S_2 \leftarrow S_1$ )
- ② Push Value into  $S_1$ . ( $S_1 \leftarrow \text{value}$ )
- ③ Pop all Elements from  $S_2$  and Push back to  $S_1$  ( $S_1 \leftarrow S_2$ )

Dequeue() :  $O(1)$

- ① Pop from  $S_1$  and return as o/p.

Let operations to be Performed are:

Enqueue(A), Enqueue(B), Dequeue(),  
Enqueue(C), Dequeue(), Enqueue(D),  
Enqueue(E), Dequeue(), Dequeue()

⇒ Expected o/p : A, B, C, D, E

Enqueue(A) :  $S_1 \mid S_2 \mid$

1.  $S_2 \leftarrow S_1$  —  $S_1 \mid S_2 \mid$

2.  $S_1 \leftarrow A$   $S_1 \mid A \mid S_2 \mid$

3.  $S_1 \leftarrow S_2$   $S_1 \mid A \mid S_2 \mid$

Enqueue(B) :  $S_1 \mid A \mid S_2 \mid$

1.  $S_2 \leftarrow S_1$   $S_1 \mid S_2 \mid A \mid$

2.  $S_1 \leftarrow B$   $S_1 \mid B \mid S_2 \mid A \mid$

3.  $S_1 \leftarrow S_2$   $S_1 \mid B \mid A \mid S_2 \mid$

Dequeue() :  $S_1 \mid A \mid B \mid S_2 \mid$

Pop from  $S_1$  and o/p : A

$S_1 \mid B \mid S_2 \mid$

Enqueue(C) :

1.  $S_1 \mid S_2 \mid B \mid$

2.  $S_1 \mid C \mid S_2 \mid B \mid$

3.  $S_1 \mid B \mid C \mid S_2 \mid$

Dequeue() :

Pop from  $S_1$  o/p : B

$S_1 \mid C \mid S_2 \mid$

Enqueue(D) :

1.  $S_1 \mid S_2 \mid C \mid$

2.  $S_1 \mid D \mid S_2 \mid C \mid$

3.  $S_1 \mid C \mid D \mid S_2 \mid$

Enqueue(E)

1.  $S_1 \mid S_2 \mid D \mid C \mid$

2.  $S_1 \mid E \mid S_2 \mid D \mid C \mid$

3.  $S_1 \mid C \mid D \mid E \mid S_2 \mid$

Dequeue() o/p : C

$S_1 \mid D \mid E \mid S_2 \mid$

Dequeue() : o/p : D

$S_1 \mid E \mid S_2 \mid$

Dequeue() : o/p : E

$S_1 \mid S_2 \mid$

obtained o/p sequence:  
A, B, C, D, E



## Topic : Queue Implementation using Stack



Dequeue() operation Costly

Enqueue() :  $O(1)$

Push value into S1.

Dequeue() :  $O(n)$

1. Pop all Elements from S1 except last Element,  
Push to S2.
2. Pop last Element from S1 and return as o/p.
3. Pop all Elements from S2 and Push back to S1

Practice Sequence of  
Enqueue() and  
Dequeue() operations.





## 2 mins Summary



- Stack Implementation Using 2 queues
- Queue Implementation Using 2 stacks

Stay Tuned . . .





**THANK - YOU**